

Introduction to Traditional knowledge (TK)

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- Traditional knowledge (TK) is originally cultured oriented and it is essential to the cultural identity of the society in which it is regulated and secured.
- Traditional Knowledge is **used from ancient times by tribal people** and by indigenous local communities under the local laws, customs and culture.
- It has been transferred and **expanded from generation to generation.**
- The role of traditional knowledge is very **important in making a country more developed.**

Introduction to Traditional knowledge (TK)

It gives a chance for traditional based literary, artistic or scientific works, performances, inventions, scientific discoveries, designs, marks, names and symbols, undisclosed information, all other traditional based inventions and creations resulting from intellectual activity.

Traditional knowledge means.....

The term Traditional Knowledge means *“the knowledge possessed by the indigenous people and communities, in one or more societies and in one or more pattern, with, but not reserved to art, dance and music, medicines and folk remedies, folk culture, biodiversity, knowledge and protection of plant varieties, handicrafts, designs, literature”*.

Importance of TK

- For indigenous people and tribal communities, the main focus is to protect their traditional knowledge and the focal point on fundamental justice and the propensity to protect and safeguard their traditions.
- In the globalized world, traditional knowledge somehow gained special importance in the past few years which have seen an outburst in the demand for herbal medicines.
- These herbal and the traditional method for making these medicine is taking an important part of human health.

Practising Traditional Knowledge

- We use **Neem** while anybody suffering from any kind of **fungal infection on skin**.
- If we cut our fingers while chopping vegetables or got hurt while playing, immediately, we **apply turmeric paste** on the wound that is healed in a day or two.
- In case of cough and cold, we use **basil leaves (tulsi)** to get rid of it.
- People, who are living in the villages, use small branches of **Neem as toothbrush**.
- In case of complaint about indigestion or stomach ache, grandma prepares **ginger or lemon decoction** which solves the problem.

Practising Traditional Knowledge

- Bone setters **set the broken bone with herbal medicines.**
- Even the **bites of** various insects including **snakes and scorpions** are cured with the use of plants and roots.
- In India, there are so many herbs which cure so many diseases. Since from ancient time, people got closer and mingled with Nature.
- Indian people understood the importance of plants and herbs in their life so they narrated it as “**Mother Nature**”.
- These are few examples of Traditional Knowledge which is practiced by various tribal and local communities of villages in India.
- This practices last from **over 5000 years.**

Practising Traditional Knowledge

- The ancient people understood the secrets of herbs, medicinal characteristics of plants.
- People are using those secrets in their day to day life.
- These developing countries are very rich in traditional type of Biological Variety.
- This life form of Biological Variety is known as “Biodiversity”.
- New products have been discovered through research which is based on “Biological Recourses” named “Bio prospecting”.

Protection of TK

- As the commercial development is increasing, technologically advanced countries are using those biological resources and biological components without taking permission or providing any fair compensation to those indigenous people.
- When these biological recourses are used in this unlawful way, it amounts to a theft called “**Bio-piracy**”.

Protection of TK

- Indian biological resources are patented in developed nations for their commercial gain which are strictly the cases of bio-piracy.
- It is the **duty of the government to protect Indian indigenous knowledge and biological resources** from commercial interest of developed nations

Protection of TK

- As we reached to 21st century, we have achieved great development and have done many inventions which can make the human life very easy, smooth and comfortable and most of such inventions have been protected under various intellectual property laws.
- But still, some communities are in existence in some parts of the world, where, the development has not reached.
- Such communities have very perfectly preserved their knowledge by passing that to their generations in heritage.

Protection of TK

- India has very rich heritage of knowledge and in ancient period, everyone had such knowledge but with development and urbanization, linkage between ancient culture and men has been disturbed and today very few people in urban areas are aware with such knowledge.
- But, some communities in villages have preserved it.
- For example, The **Fish Therapy for “asthama”** is exclusive knowledge of some communities of Hyderabad.
- Thus, this kind of knowledge which is preserved by some communities is considered as Traditional Knowledge.

Definition of TK

- There is **no agreed definition** for “traditional knowledge”.
- The word Traditional Knowledge is itself a suggestive word.
- **Traditional** means **hereditary** or which is given by generation to generation.
- **Knowledge** means **useful information**.
- Thus Traditional Knowledge refers to that **useful information which is passes through one generation to other generation.**

Definition of TK

- Thus, Traditional Knowledge refers to knowledge, innovations and practice of indigenous and local communities around the world.
- This is the knowledge which has been preserved in **human memories and not written anywhere.**
- It is accessible **through recall and practice** of learned skills in a useful way in day to day life.

IMPORTANT CHARACTERISTICS OF TK

Some important characteristics of TK can be identified as follows:

- It is **transmitted from generations to generations** in many cases.
- It is **transmitted orally** for generations from person to person.
- It is being considered by the communities **as gift of God** and not as a private property.
- It is usually **impossible to identify the original creator** of the information.

IMPORTANT CHARACTERISTICS OF TK

- It is learned through continuous observation, experience and practice.
- It is inseparable part of communal and cultural life of its holders, and
- It is usually associated with the biological resources.
- It is not limited to any specific field of technology or the arts.
- It is owned by a community and its use is often restricted to certain members of that community.

SCOPE AND IMPORTANCE OF TK

- Traditional and indigenous knowledge has been used for centuries by indigenous and local communities under local laws, customs and traditions.
- TK has played, and still plays, an important role in vital areas such as medical treatment, food security and the development of agriculture.
- TK is also the cause of a great variety of artistic expressions, including musical works and handicrafts.

SCOPE AND IMPORTANCE OF TK

- TK is a central component for the daily life of millions of people in developing countries.
- Traditional Medicine (TM) serves the health needs of a vast majority of people in developing countries where access to modern health care services and medicine is limited by economic and cultural reasons.
- It is the only affordable treatment available to poor people and in remote communities.

SCOPE AND IMPORTANCE OF TK

- TK constitutes the ancient knowledge of humanity, the deepest layer on which our science and culture have developed, the local solutions that have allowed the creation and management of ecosystems and cultural landscapes on the entire surface of the planet.
- It enables the development of solutions with a low energy and resource use that are able to adapt to environmental variability and to react to emergencies and catastrophes in flexible and multifunctional ways.

SCOPE AND IMPORTANCE OF TK

- Today, while entire planet systems risk ecological collapse shows how to interact with the environment enhancing its resource potential without exhausting it.
- In addition, it cannot be excluded that traditional knowledge might have an industrial application, even if the tangible object to which the intangible knowledge relates has not been subject to any scientific interference or modification.
- TK is thus a valuable source of knowledge. TK may help to find useful solutions to current problems, sometimes in combination with modern scientific and technological knowledge.

Importance of Traditional Knowledge

- Traditional Knowledge plays a key role in the preservation and sustainability of diversity.
- Many activities based on traditional knowledge are important sources of income, food and healthcare for the indigenous people.
- The long term economic development of many indigenous and local communities, depend on their ability to harness their traditional knowledge for commercial benefits.
- Traditional Knowledge is being rapidly lost as local ecosystems are degraded and traditional communities are integrated into the wider society.

TKS – Nature and Type

- The evolution of TKS is very much local in nature and associated with a particular environmental and/or socio-cultural context.
- Therefore, TKS has the characteristics of local, empirical, time tested dynamisms.
- Moreover, TKS is always handed over or transferred from one generation to another and also between communities mostly orally and/or visually.

Nature and types of TK

Traditional Knowledge is divided into

- Cultural Knowledge
- Artistic Knowledge
- Medicinal Knowledge
- Biodiversity/ Natural Resources Knowledge
- Agricultural Knowledge
- Sacred Knowledge

Nature and types of TK

From its domain of application and associated management approaches, TKS can be categorised as

- (i) Traditional Ecological Knowledge (TEK),
- (ii) Traditional Technical Knowledge (TTK) and
- (iii) Traditional Value and Ethics (TVE).

Traditional Ecological Knowledge (TEK)

- TEK refers to the evolving knowledge acquired by indigenous and local people over hundreds or thousands of years through direct contact with the environment.
- This knowledge is specific to a location and includes the relationships between plants, animals, natural phenomena, and the landscape that are used for livelihood.

Traditional Technical Knowledge (TTK)

- TTK represents the knowledge related to design and development of tools, implements and gears for different application in the context of natural resource management by the indigenous communities.
- Such practices are related to agriculture, fisheries, animal husbandry, forestry, handloom and handicraft etc.
- Moreover, TTK also represents the knowledge and skill about design and construction like housing, water harvesting structure, roads and bridges, etc.

Handlooms and handcrafts



Traditional Value and Ethics (TVE)

- During the process, it evolves the concept of sacred species, space, forests, water bodies, etc.
- This involves seasonality based practices like restriction of fishing during breeding season, harvesting forest resources during flowering period, etc.

Traditional Water harvesting practices

- There are many age-old-practices of harvesting water in the country.
- Basically to collect rainwater, restore surface flow of water, ground water recharging, etc.
- These are based on simple technology and defined management principles.

Step well, an ancient water harvesting structure

- A step well is exactly what it sounds like- steps down to a well.
- The earliest step wells date back to about 550 AD were developed in India as a necessity for areas suffering from torrential seasonal rains



TANKA, AN ANCIENT WATER HARVESTING SYSTEM @RAJASHTHAN



JOHAD, AN ANCIENT PRACTICE OF RAJASTHAN



ZABO A TRADITIONAL PRACTICES AMONG THE NAGA COMMUNITIES, NAGALAND.



Bamboo drip irrigation

- In different states of North-Eastern part of India Bamboo drip irrigation is a common practice.
- The design of the Bamboo pipe for irrigation varies with variation of rainfall, which reflects the uniqueness of traditional knowledge system of the local communities

**BAMBOO DRIP
IRRIGATION OF KARBI-
ANGLONG, ASSAM**



**BAMBOO DRIP
IRRIGATION
OF MEGHALAYA**



Traditional Housing: Vernacular Architecture

- Usually these are called Vernacular Architecture, which is an architectural style and design based on local needs, availability of construction materials and reflecting local traditions.



VERNACULAR HOUSING OF KERALA



Traditional agricultural practices

- These practices are followed in selection of crop varieties, land selection, land preparation, soil fertility, management, pest and disease management, irrigation, harvesting, post-harvest management, seed preservation, etc.
- Moreover there are different tools and implements used for the purposes

Traditional agricultural practices

- There are different shapes and sizes of ploughs and hoes used for tilling of soil in the country, which varies from region to region based on soil quality, terrain condition and the crop used for cultivation.



TRADITIONAL PLOUGHING IN INDIA



Seed Preservation Technique

- In different regions people adopt different methods of seeds preservation and storage.



Seed Preservation Technique



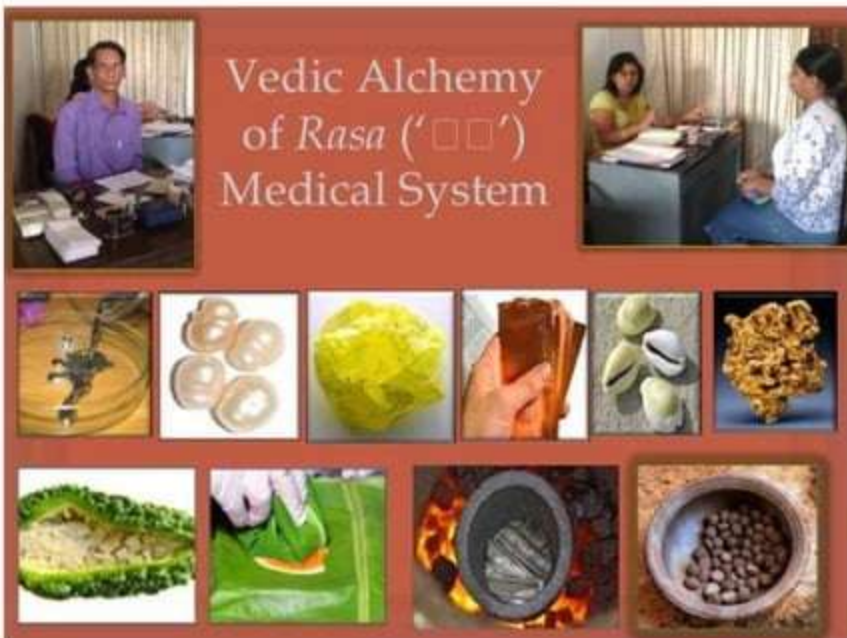
Weather Forecasting/ Prediction

- There are many methods of weather prediction practiced by the farmers in different parts of the country.
- For example, farmers in Himachal Pradesh believe that if the **honeybee flies toward northern hill** there will be no rainfall, if they **fly towards south there will be good rainfall.**
- On the other hand, in Rajasthan many local communities believe that **appearance of many butterflies together indicate a good rain and get better crops.**

Weather Forecasting/ Prediction



Traditional knowledge in health and well being



Traditional knowledge in health and well being

Ayurveda



What is Ayurveda?

➤ **Ayurveda** is the ancient Indian system of natural and holistic medicine. When translated from Sanskrit, Ayurveda means “**the science of life**” (the Sanskrit root **ayur** means “**longevity**” or “**life**” and **veda** means “**science**”).

➤ The knowledge of Ayurveda was **passed on orally** through a lineage of sages in India until it was collated into text **more than five thousand years ago**.

➤ The oldest known texts on Ayurveda are the **Charaka Samhita**, **Sushruta Samhita**, and the **Ashtanga Hrudaya**.

Traditional knowledge in health and well being

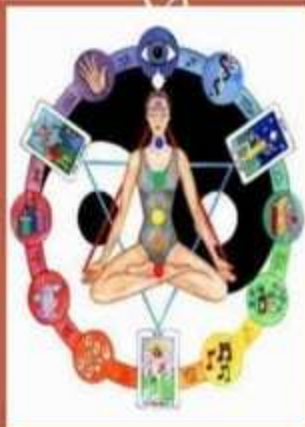
*Ass Vedakama (Eye
Treatments)*



Traditional knowledge in health and well being

Homeopathy is a medical system based on the belief that the body can cure itself. Those who practice it use tiny amounts of natural substances, like plants and minerals. **They** believe these stimulate the healing process. It was developed in the late 1700s in **Germany.**

Homeopathy

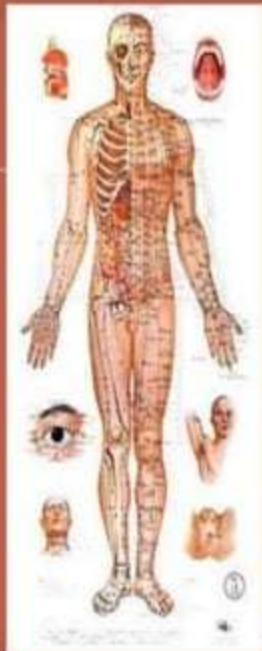


Traditional knowledge in health and well being

By stimulating the body's various systems, acupuncture can help to resolve **pain**, and improve sleep, digestive function, and sense of well-being.



Acupuncture



- Thank you

